



BAHIR DAR UNIVERSITY

BAHIR DAR INSTITUTE OF TECHNOLOGY

Faculty of Computing Department of Software Engineering

TITLE=OPENINDIANA OS



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1.. Introduction (Background and Motivation)

Operating systems are essential software that manage computer hardware and provide services for users and applications. They make it possible to run programs, manage files, and control system resources efficiently. Different operating systems are designed with different goals such as performance, stability, and security.

One of the operating systems used in this project is OpenIndiana, which is a Unix-based operating system derived from OpenSolaris. It is known for its stability, advanced filesystem support like ZFS, and strong performance in system-level operations. Although it is not as commonly used as Windows or Linux, it is important for understanding UNIX-based systems and their architecture.

The development of virtualization technology has made it easier to experiment with different operating systems without affecting the main computer. Using tools like VirtualBox or VMware, it is possible to install and run OpenIndiana in a virtual environment. This approach is safe and flexible, especially for students who want to practice installation and system configuration.

The main motivation of this project is to gain practical knowledge about installing and working with OpenIndiana in a virtual machine. It helps to understand how operating systems function in real situations, not just in theory. During the process, different challenges may occur, and solving them improves problem-solving skills and technical confidence.

2.. Objectives

The main objective of this project is to understand and work with the OpenIndiana operating system in a practical way using a virtual environment.

Specifically, this project aims to install OpenIndiana on virtualization tools such as VirtualBox or VMware and learn the step-by-step installation process. It also focuses on understanding the basic features of OpenIndiana, including its system structure, performance, and file system support.

Another objective is to gain hands-on experience in managing and configuring the operating system. This includes creating user accounts, exploring system commands, and understanding how resources are managed within OpenIndiana.

In addition, the project aims to identify common problems that may occur during installation or usage and find appropriate solutions. This helps to improve problem-solving skills and builds confidence in handling real system issues.

Finally, the project helps to develop a deeper understanding of UNIX-based operating systems through OpenIndiana and to learn how virtualization makes it easier to test and use different operating systems safely.

3. Requirements

To successfully complete this project using OpenIndiana, both hardware and software requirements are needed. These requirements are important to make sure the installation process runs smoothly and the system works without problems.

First, for the hardware requirements, a computer or laptop with a good performance is necessary. The system should have at least a dual-core processor so that it can handle both the main operating system and the virtual machine at the same time. In addition, a minimum of 4GB RAM is required, but having 8GB RAM is better for smoother performance, especially when running OpenIndiana inside a virtual environment.

Storage is also important, and at least 20GB of free disk space should be available to install the operating system and store related files. If the storage is too small, the system may become slow or fail during installation. A stable and well-functioning computer is also needed to avoid unexpected shutdowns or errors.

Secondly, for the software requirements, a virtualization tool is required to install OpenIndiana. This project uses software like Oracle VM VirtualBox or VMware Workstation to create a virtual machine. These tools allow us to run OpenIndiana as a guest operating system without affecting the main system. The OpenIndiana ISO file is also required, which is the installation file used to set up the operating system inside the virtual machine.

Moreover, the host operating system, such as Windows or Linux, must already be installed on the computer to support the virtualization software. It is also important to make sure that virtualization is enabled in the BIOS settings of the computer. Without enabling this feature, the virtual machine may not run properly. Basic system updates and drivers should also be installed to avoid compatibility issues during the setup.

Overall, having the correct hardware and software requirements ensures that OpenIndiana can be installed and used effectively in a virtual environment without major problems.

4. Installation process

In this project, OpenIndiana is installed using a virtual environment. The installation is done using a virtualization tool to avoid affecting the main operating system.

Step 1: Install Virtualization Software

VirtualBox

[Home](#) [Download](#) [Documentation](#) [Community](#) 

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Change Log

[List of changes](#)

File Checksums

[SHA256 checksums](#) [MD5 checksums](#)

User Guide

[User Guide \(PDF\)](#) [User Guide \(HTML\)](#)

VirtualBox SDK

VirtualBox 7.2.6 Software Developer Kit (SDK)

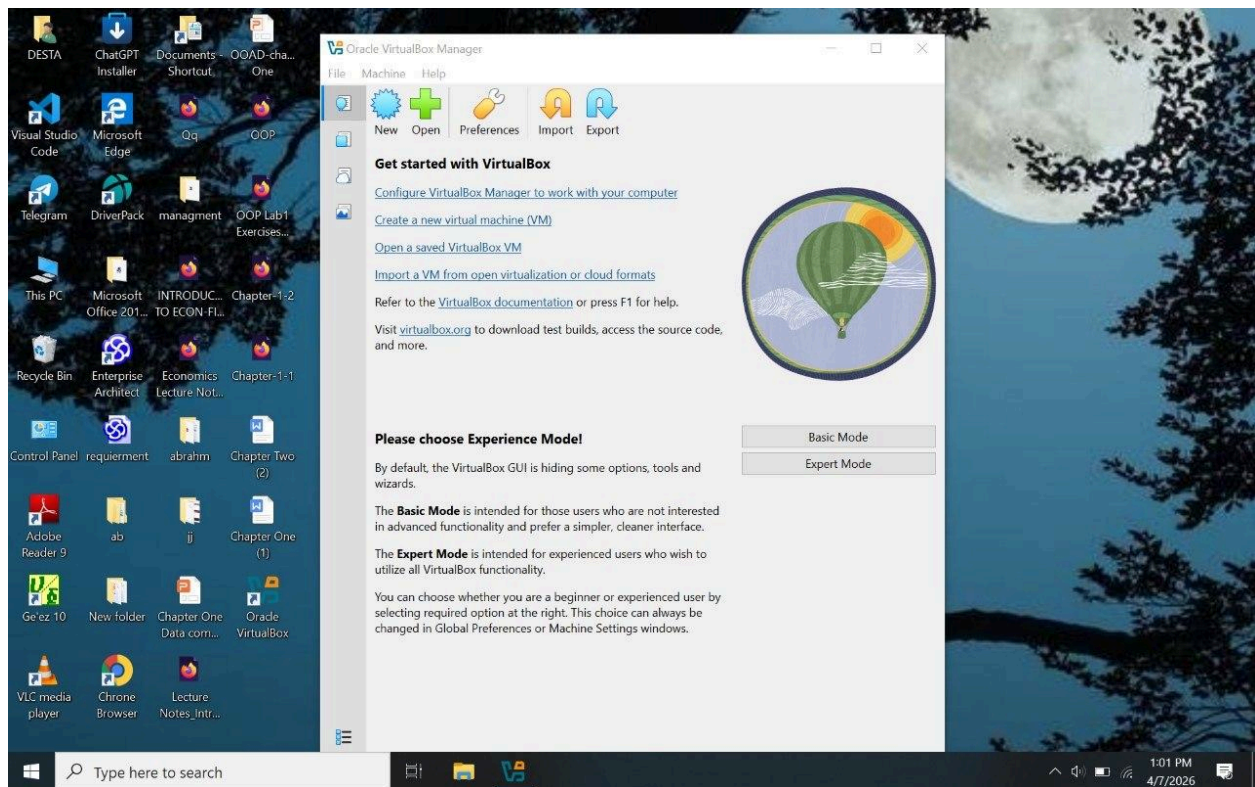
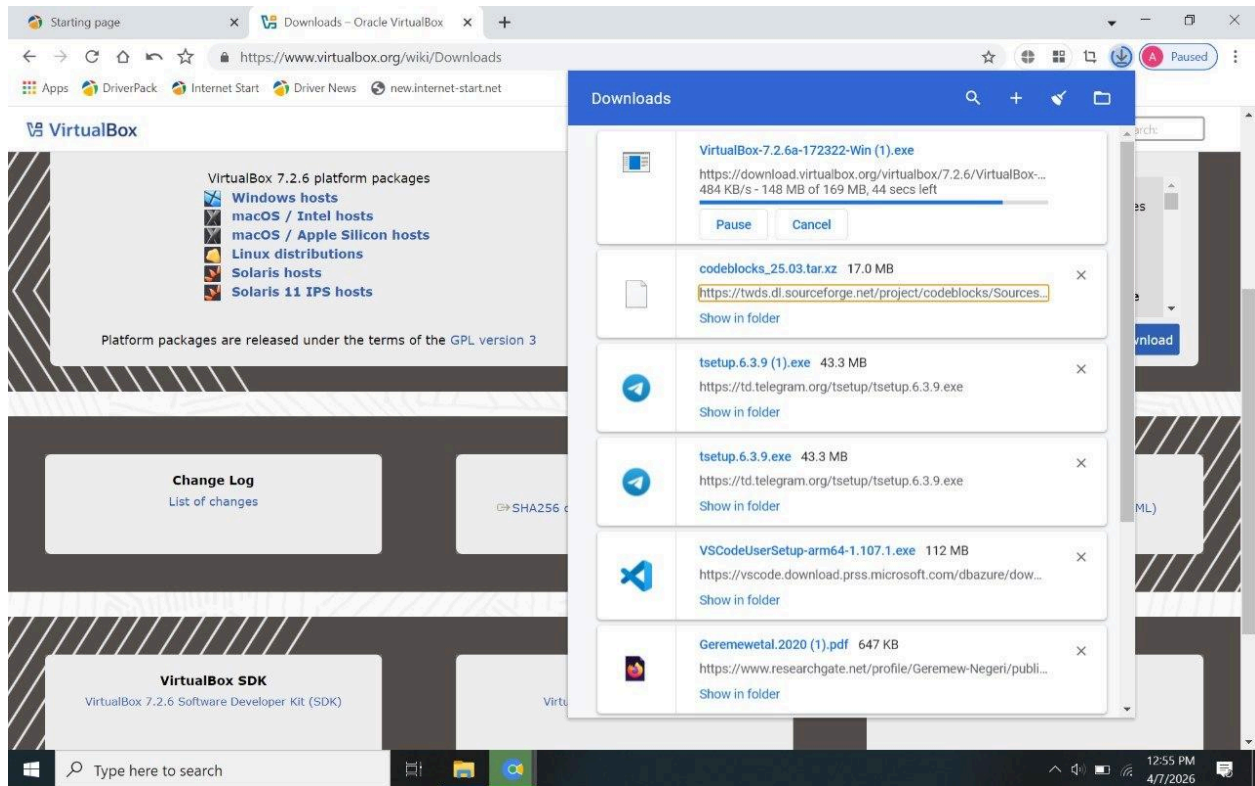
Source Code

VirtualBox 7.2.6 Source Code

Previous Releases

VirtualBox older builds

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Step 2:Download OpenIndiana ISO File



Download

Getting started

- [How to Install OpenIndiana](#)
- [How to Upgrade OpenIndiana](#)

OpenIndiana works on most x86 PCs ([SPARC has Beta ISO's, support Welcome](#)).

From 2017.04 onwards, running on 32-bit systems is not supported anymore and the userland is being migrated progressively to 64-bit only. As usual, 32-bit libraries and core applications are still delivered and maintained for the sake of backward compatibility (but only when running a 64-bit kernel).

To check if your hardware is supported, you can refer to the [illumos HCL](#) and the [OpenIndiana Community HCL](#); alternatively, you can boot a Live install media and run the Device Driver Utility (ddu).

For instructions regarding downloading, preparing and installing the images, please refer to the [OpenIndiana Handbook](#).

If you want to contribute to the project, [find out how you can help](#).

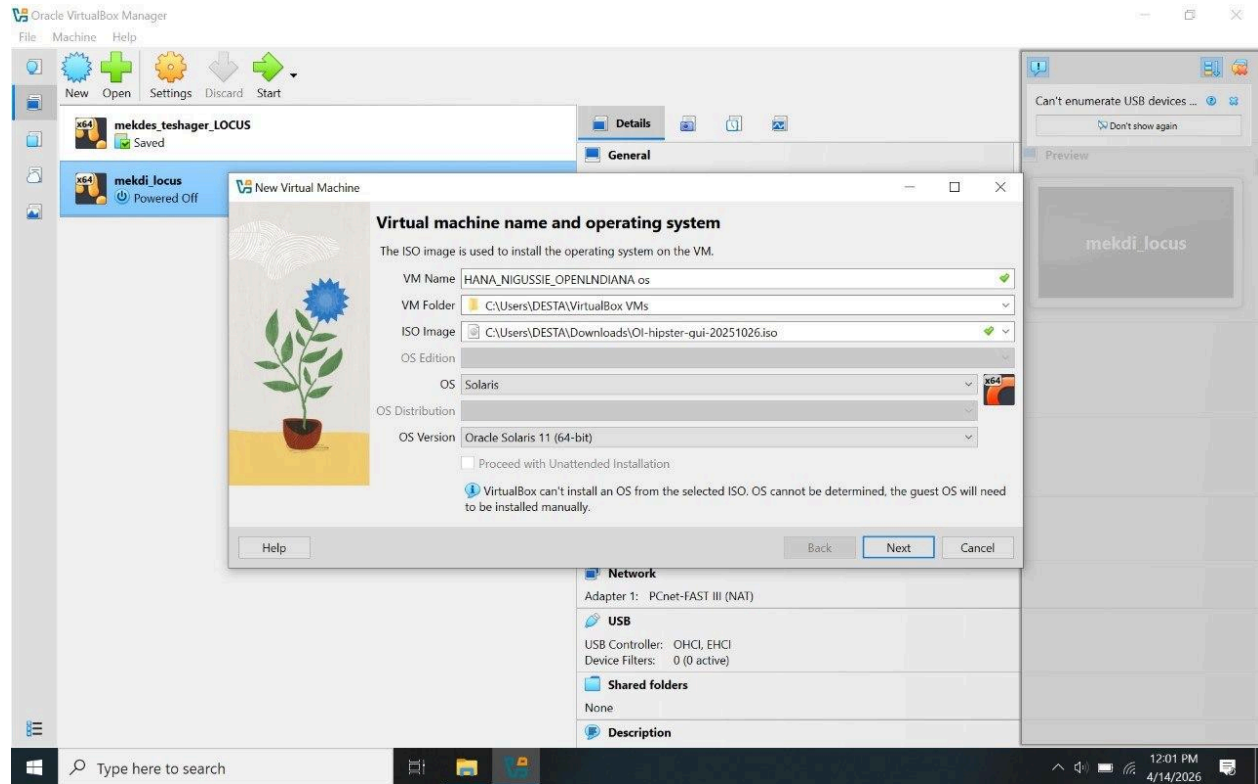
Download the current Hipster snapshot of OpenIndiana.

ISO	Size	Checksum
OpenIndiana Hipster 2025.10 Live DVD (64-bit x86)	1.90G	sha256
OpenIndiana Hipster 2025.10 Live USB (64-bit x86)	2.30G	sha256
OpenIndiana Hipster 2025.10 Text Install DVD (64-bit x86)	970.3M	sha256
OpenIndiana Hipster 2025.10 Text Install USB (64-bit x86)	1.20G	sha256
OpenIndiana Hipster 2025.10 Minimal Install DVD (64-bit x86)	470M	sha256
OpenIndiana Hipster 2025.10 Minimal Install USB (64-bit x86)	604M	sha256
OpenIndiana Hipster 2025.10 CloudImage (64-bit x86)	831M	sha256

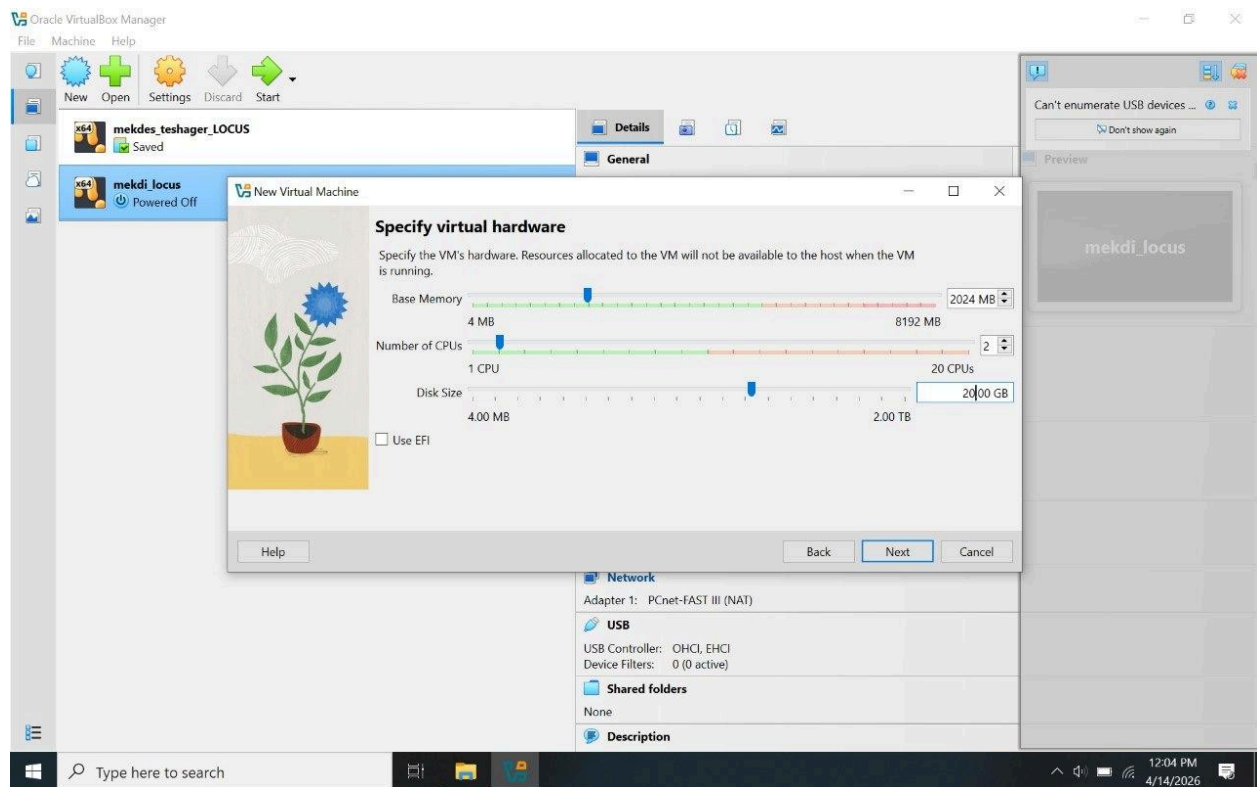
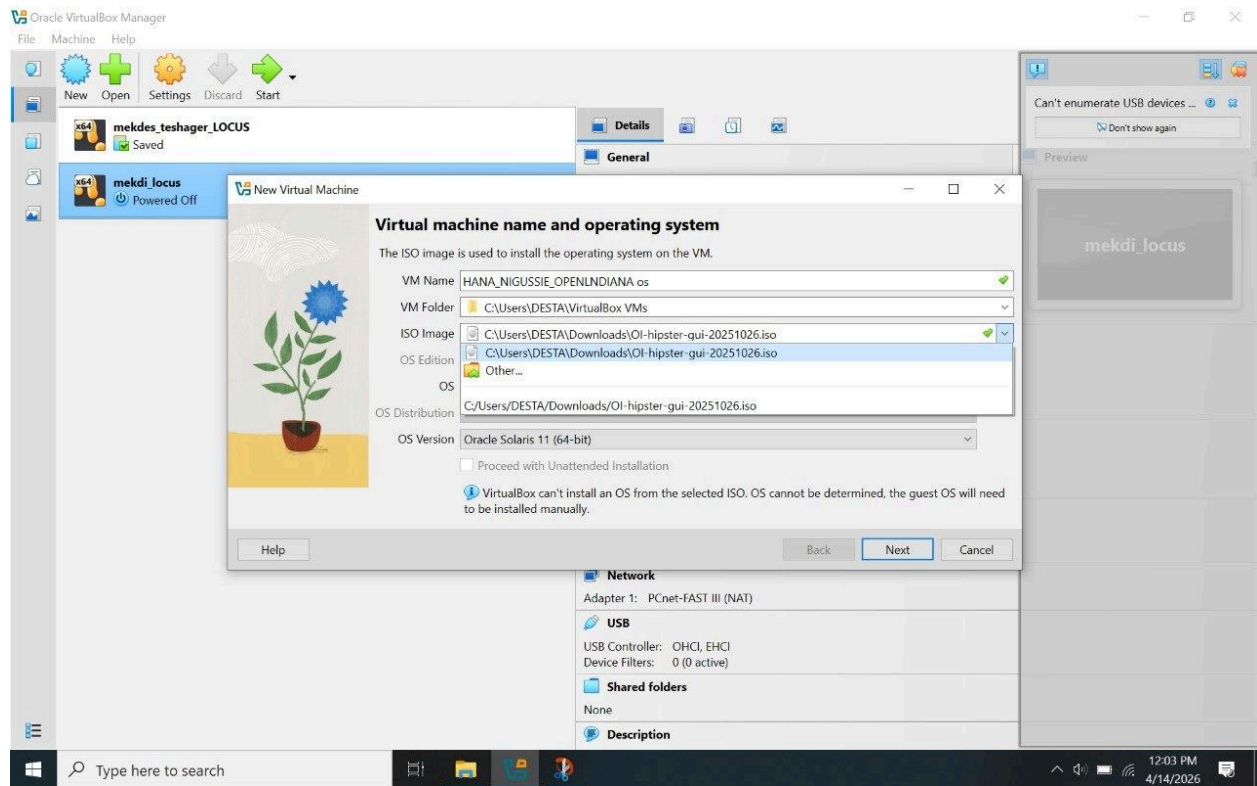
Please note that these links might be slower, and it is recommended to visit a list with [mirror sites in your region](#). For more mirrors, our documentation has a list of [alternate download sites](#).

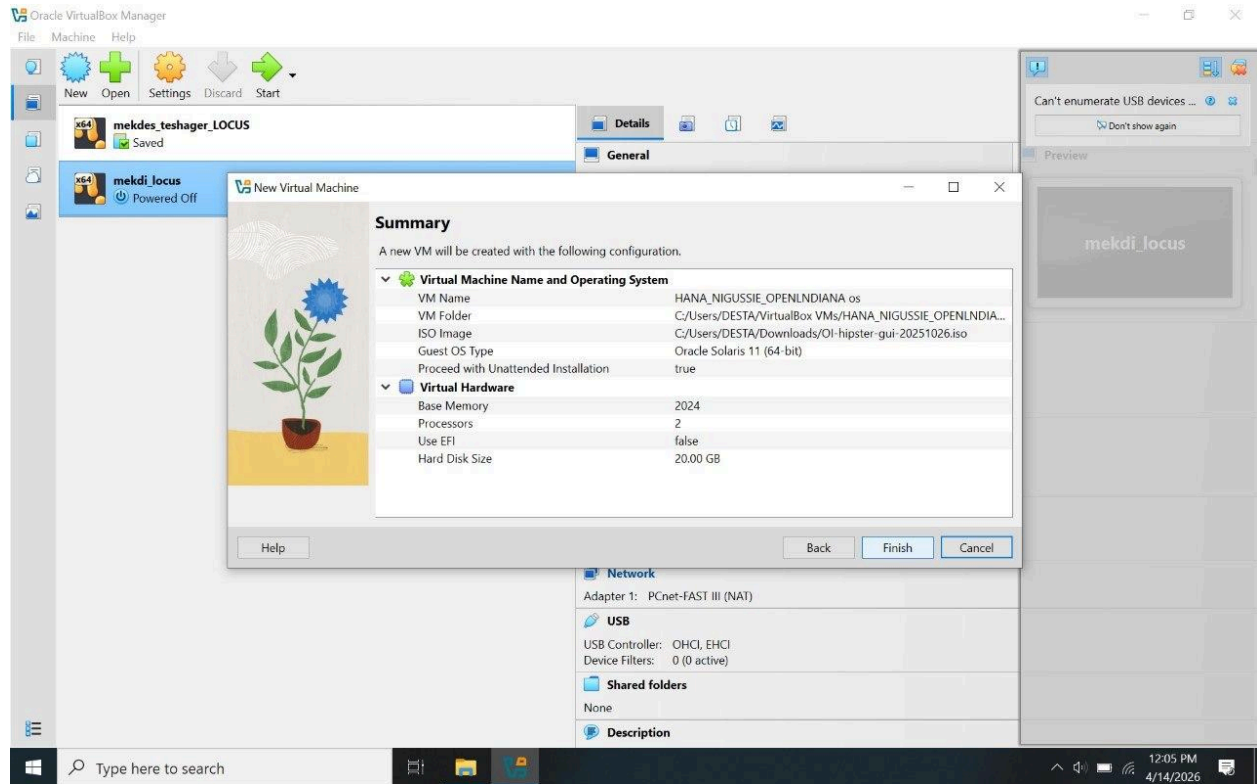
These installations can be updated regularly using the Hipster repository, and receive security fixes. Starting with the 2017.04 release, every image is signed with OpenIndiana Release Engineering GPG key id DBE31887.

Step 3: Create a New Virtual Machine

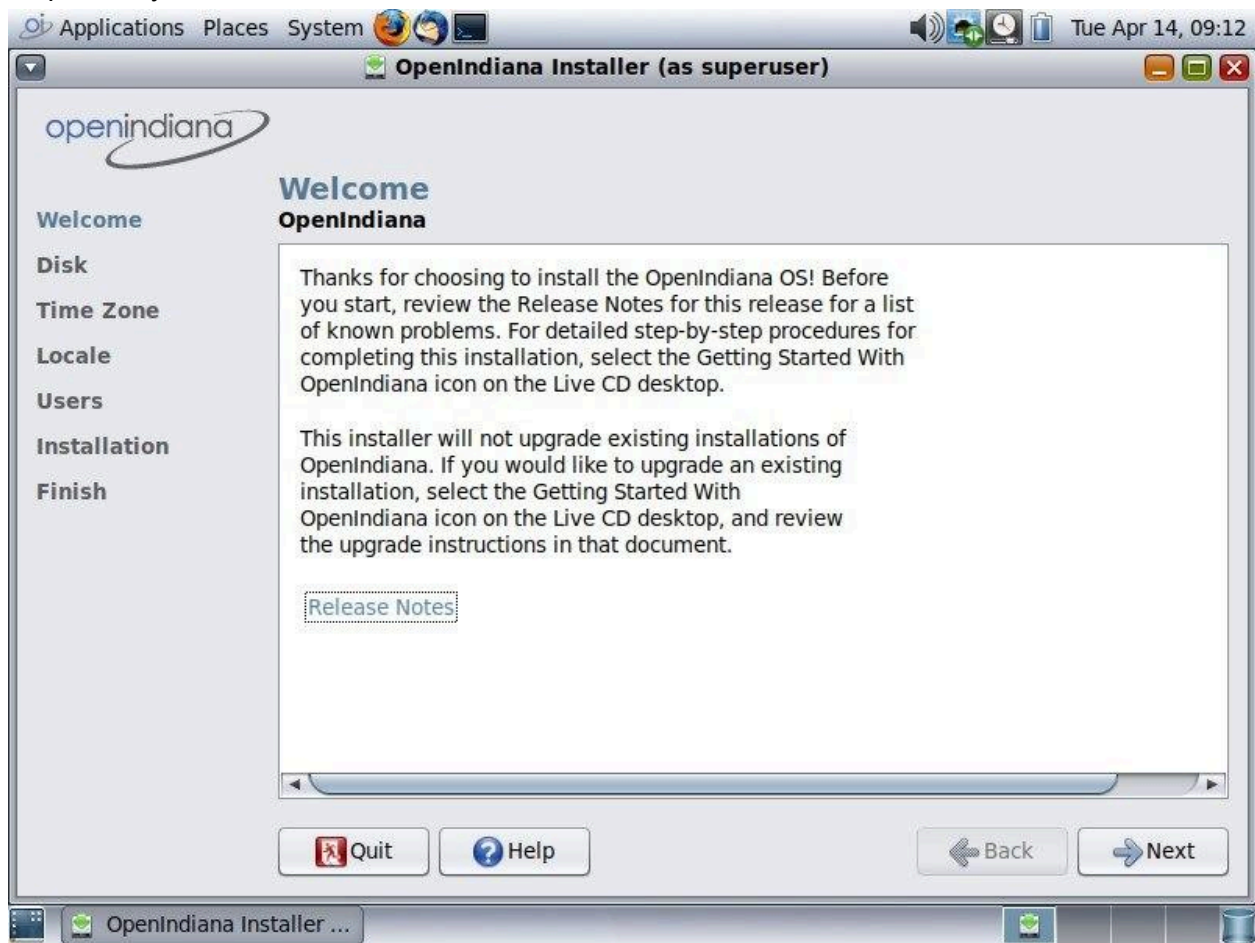


Step 4:allocate memory or RAM and hard disk and attach openIndiana image file

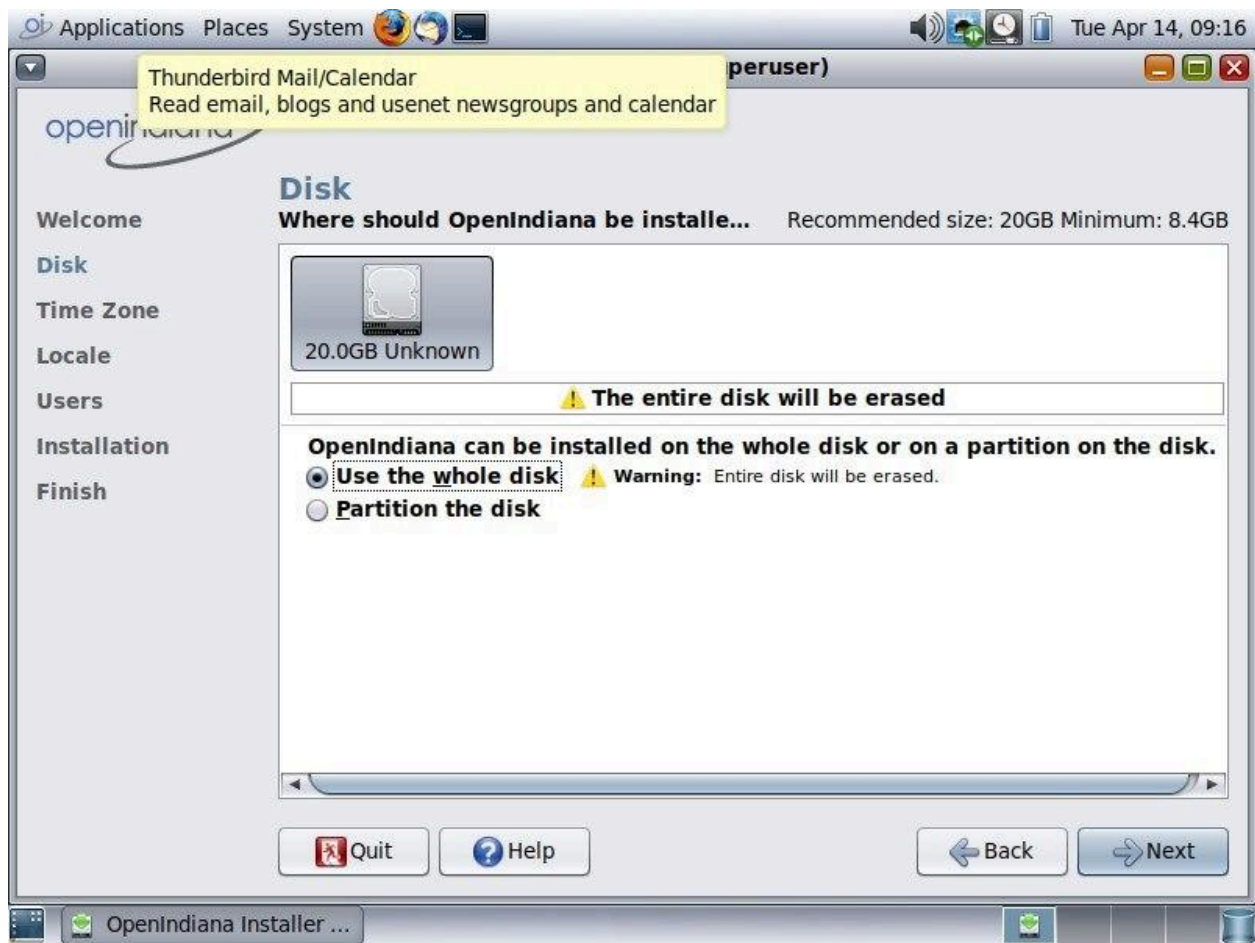




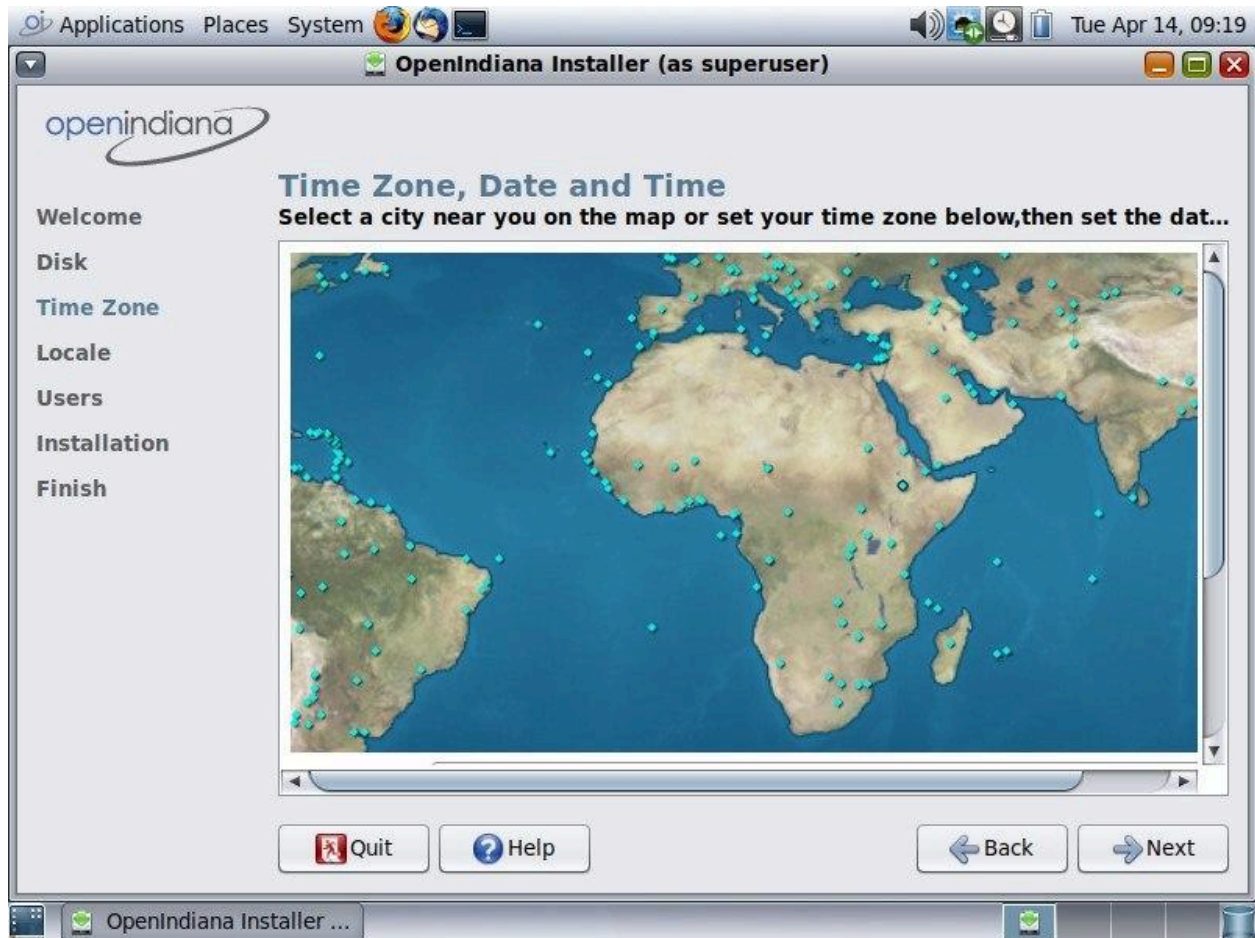
Step 5: it says welcome



Step 6: then allocate the disk



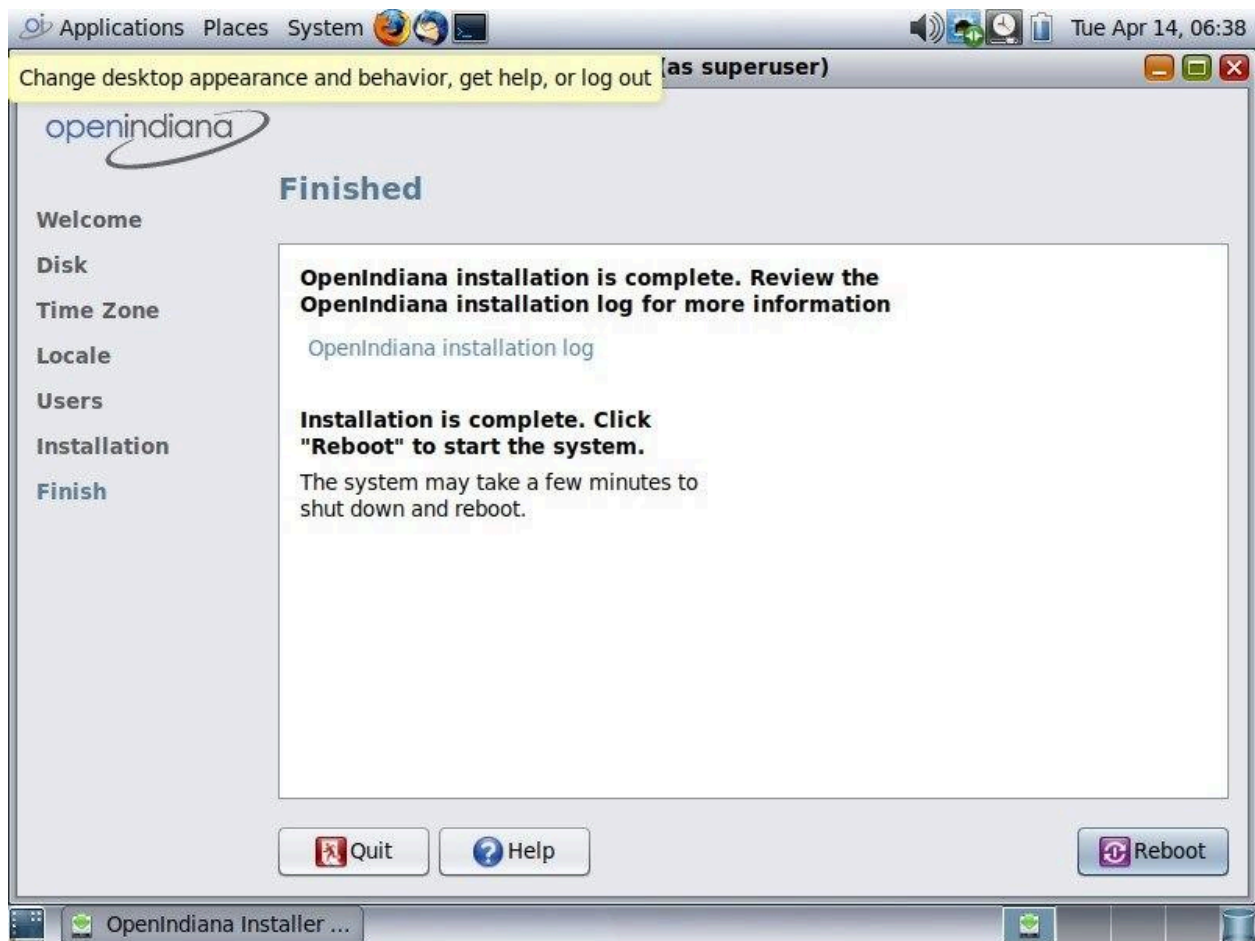
Step 7:next prepare the timezone and locale



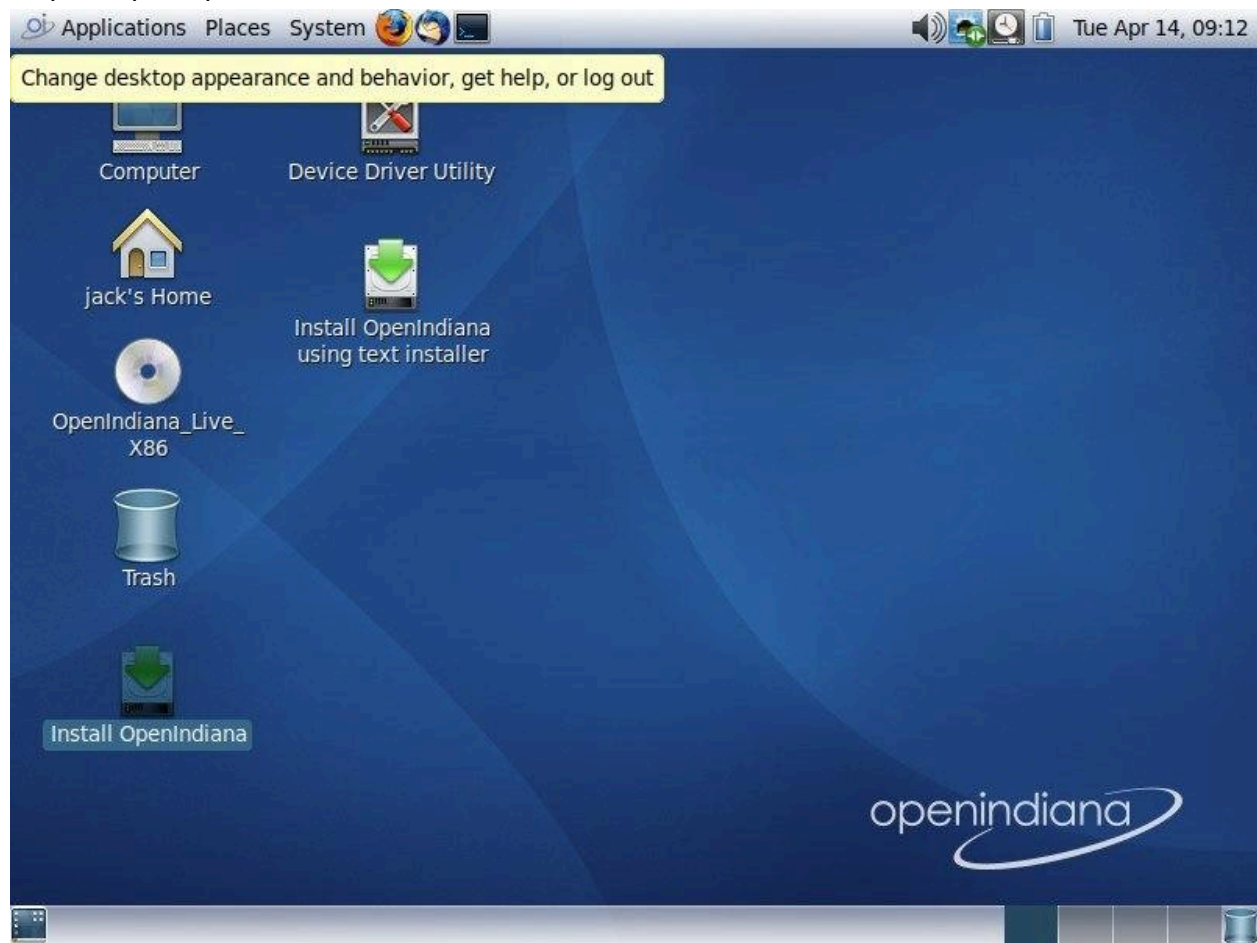
Step 8:set password and user name



Step 9:next do the instalization and finish



Step 10:open openindiana



Step 11:select install openIndiana and install it



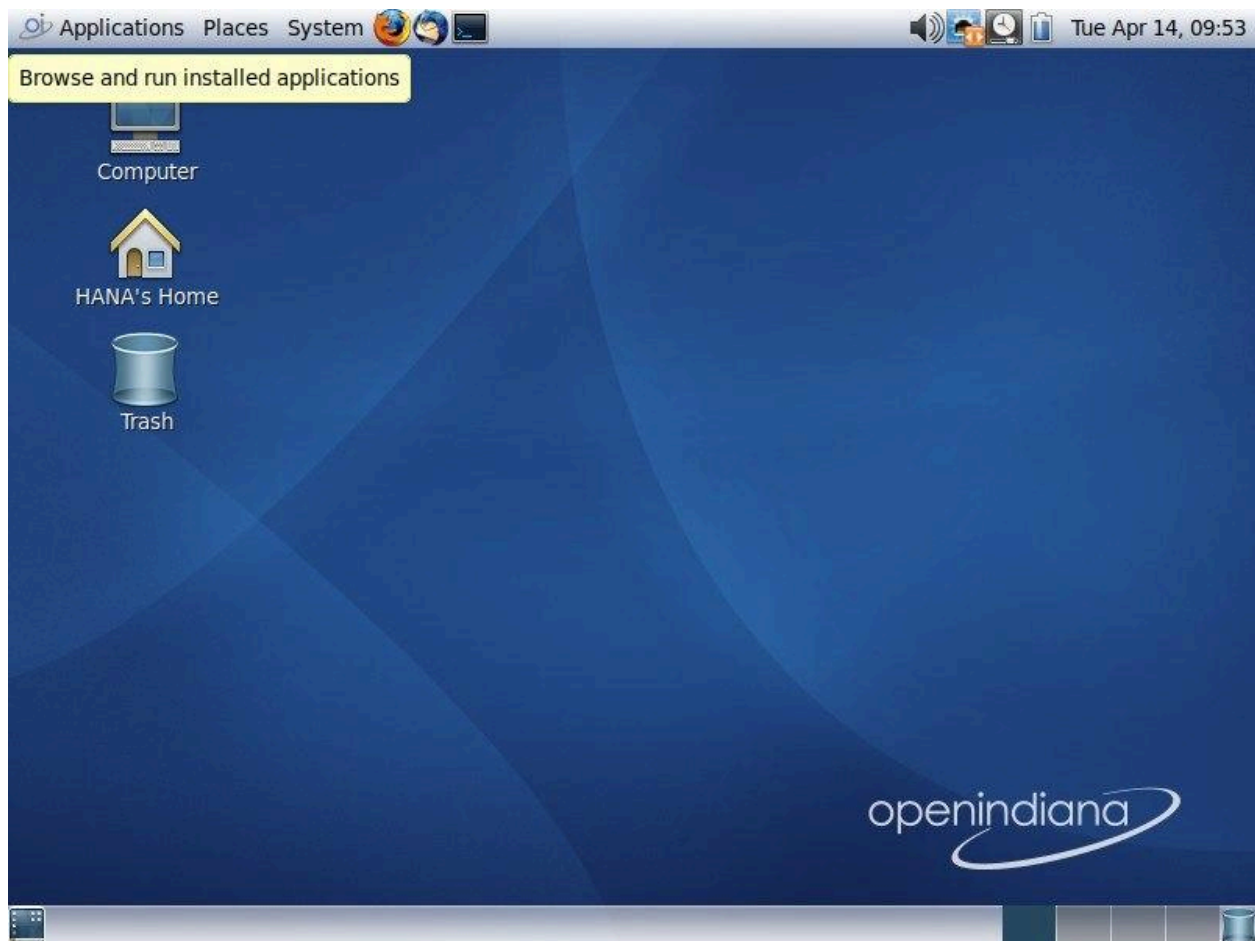
```
openindiana

Loading unix...
Loading /platform/i86pc/amd64/boot_archive...
Loading /platform/i86pc/amd64/boot_archive.hash...
Booting...
OpenIndiana Hipster 2025.10 Version illumos-68259130da 64-bit
Loading NVIDIA Kernel Mode Setting Driver for UNIX platforms 470.256.02 Thu May 2 14:30:39 UTC
2024
Hostname: openindiana
Remounting root read/write
Configuring devices.
Probing for device nodes ...
Preparing live image for use
Done mounting Live image

1. Arabic                12. Hungarian
2. Catalan              13. Indonesian
3. Chinese - Simplified 14. Italian
4. Chinese - Traditional 15. Japanese
5. Czech                16. Korean
6. Dutch                17. Polish
7. English              18. Portuguese - Brazil
8. French               19. Russian
9. German               20. Slovak
10. Greek               21. Spanish
11. Hebrew              22. Swedish

To select the language you wish to use, enter a number [default is 7]:
```

Step 12: use my password and username and get the openindiana



5.Issues (problem faced) and solutions

1.slow performace

During the installation and even after successfully installing OpenIndiana, I noticed that the system performance was very slow. Simple tasks like opening applications or navigating through the system took a long time. Sometimes the system even lagged when switching between windows.

Solution:

After observing the problem, I understood that the main cause was insufficient resource allocation in the virtual machine. Initially, I had assigned only 2GB of RAM and one CPU core, which was not enough for OpenIndiana to run smoothly. To solve this, I increased the RAM to 4GB and assigned more CPU cores from the settings of Oracle VM VirtualBox.

2.Installation Freezing or Taking Too Long

Another problem I faced was during the installation process. At some point, the installation either froze or took a very long time without showing progress. This made it difficult to know whether the system was still working or not.

Solution:

To solve this issue, I first checked whether my computer had enough free disk space and system resources. I realized that my system was running many background applications, which affected the installation speed. I closed all unnecessary programs and restarted the installation process.

3. Network Connection Not Working

After installing OpenIndiana, I faced a problem where the internet connection was not working inside the virtual machine. This made it difficult to download updates or test network-related features.

Solution:

To fix this problem, I checked the network settings of the virtual machine. Initially, the network adapter was not properly configured. I changed the network mode to “NAT” in Oracle VM VirtualBox, which allows the virtual machine to share the host system’s internet connection.

4. Difficulty in Finding the Terminal

While using OpenIndiana, I had difficulty finding the terminal application. Unlike other operating systems like Linux where the terminal is easily accessible, in OpenIndiana it was not immediately visible, and this made it confusing at the beginning.

Solution:

To solve this problem, I explored the system menu and searched for the terminal manually. After some navigation, I found the terminal application under the system tools section. I also learned that I can use the search option to quickly locate applications.

6. File system support

In OpenIndiana, the main filesystem used is ZFS (Zettabyte File System). This is one of the most advanced and powerful filesystems compared to traditional ones like NTFS or FAT32. ZFS is designed to provide high performance, strong data protection, and easy management. Unlike older filesystems, it combines both the filesystem and volume management into a single system. This makes it easier to handle storage without needing separate tools.

One of the most important features of ZFS is data integrity. It uses a method called checksumming to detect and fix data corruption automatically. This means that if any data becomes damaged, the system can identify the problem and recover it, which improves reliability.

Another important feature is snapshot support. ZFS allows users to take snapshots of the system at a specific time. This is very useful because if something goes wrong, the system can be restored to a previous state without losing important data.

ZFS also supports large storage capacity and can handle a huge amount of data efficiently. It is more scalable compared to traditional filesystems like FAT32, which have limitations in file size and partition size.

In addition, ZFS uses a feature called copy-on-write. This means that instead of overwriting existing data, it writes new data in a different location. This reduces the risk of data loss and improves system stability.

Although OpenIndiana mainly uses ZFS, it can also support other filesystems like UFS (Unix File System), but ZFS is the default and most recommended because of its advanced features.

7. Advantages and disadvantages

Advantages:

- OpenIndiana uses ZFS filesystem, which provides strong data protection and supports large storage
- High system stability and reliable performance
- Good security features for protecting data and managing users
- Supports virtualization (can run easily on tools like Oracle VM VirtualBox)
- Useful for learning UNIX-based operating systems
- Supports advanced features like snapshots and data recovery

Disadvantages

- Not very user-friendly for beginners
- Limited software/application support
- Hardware compatibility issues with some devices
- Less community and support compared to Windows and Linux
- Difficult to find solutions sometimes when problems occur
- Not widely used in everyday environments

8. Conclusion

In this project, I was able to successfully install and work with OpenIndiana in a virtual environment. This helped me to understand how operating systems are installed and managed in real situations, not only in theory.

Through this process, I learned about virtualization, system configuration, and how to solve different problems that occurred during installation and usage. I also gained experience in using system tools and understanding how resources like memory and storage are managed.

One of the most important things I learned is the role of advanced filesystems like ZFS, which provides better data protection and system reliability. Even though OpenIndiana is not widely used, it is very useful for understanding UNIX-based systems and improving technical skills.

Overall, this project helped me to build confidence in working with operating systems and improved my problem-solving ability.

9.Future outlook and recommendation

- OpenIndiana can be improved by making the user interface more user-friendly
- More software support should be added to make it more practical for daily use
- Better hardware compatibility would help users install it easily on different systems
- It is recommended for students to practice OpenIndiana using virtualization tools for better understanding
- Future users should allocate enough system resources (RAM, CPU, storage) to avoid performance issues
- Learning UNIX-based systems like OpenIndiana is important for software engineering students

Virtualization in modern OS (What, Why, and How)

What is virtualization?

Virtualization is a technology that allows one physical computer to run multiple operating systems at the same time. It creates a virtual environment called a virtual machine (VM), where another operating system like OpenIndiana can be installed and used just like a real computer.

In this project, virtualization was used to install OpenIndiana inside a virtual machine instead of installing it directly on the physical system.

Why virtualization is important?

Virtualization is very important in modern operating systems for several reasons:

It allows users to run different operating systems on one computer without affecting the main system

It provides a safe environment for testing and experimenting (for example, installing OpenIndiana without risk)

It helps save cost because there is no need for multiple physical machines

It is useful for learning and practice, especially for students in software engineering

It improves resource utilization by sharing hardware between multiple systems

How virtualization works?

Virtualization works using special software such as Oracle VM VirtualBox or VMware Workstation. These tools create virtual machines that act like real computers.

The virtualization software uses the physical computer's resources (CPU, RAM, storage) and divides them between the host operating system and the virtual machine. Each virtual machine runs independently and has its own operating system, such as OpenIndiana.

In addition, the system uses a component called a hypervisor, which manages the virtual machines and controls how resources are shared. This ensures that each virtual machine runs smoothly without interfering with others.

Relation to OpenIndiana

In this project, virtualization made it possible to install and use OpenIndiana safely. It allowed me to practice installation, test features, and solve problems without damaging the main operating system. This shows how virtualization is very useful in modern computing.

10.UNIX Standardization and mplementation (related to OpenIndiana)

UNIX standardization is important to ensure that different UNIX-based systems follow common rules and can work together. Since OpenIndiana is a UNIX-based operating system, it follows many of these standards to maintain compatibility and stability.

ISO C

- International Organization for Standardization defines the standard for the C programming language
- OpenIndiana uses C for system-level programming
- This helps programs written in C run correctly on OpenIndiana and other UNIX systems

IEEE POSIX

- Institute of Electrical and Electronics Engineers developed POSIX standards
- OpenIndiana follows POSIX standards for system calls and interfaces

- This allows software developed for other UNIX systems to run on OpenIndiana with minimal changes

FIPS

- Federal Information Processing Standards ensures security and compatibility
- OpenIndiana can support systems that require secure and standardized environments
- Useful in systems where data protection is important

Unix system implementation

- OpenIndiana is based on UNIX system design principles
- It includes components like kernel, file system (ZFS), and system calls
- Supports multitasking and multi-user environment

Unix System Release V4 (SVR4)

- OpenIndiana is indirectly based on System V architecture
- SVR4 influenced the development of Solaris, which later led to OpenIndiana
- This provides strong system performance and reliability

4.4 BSD

- Some features in OpenIndiana are influenced by BSD systems
- Especially in networking and system utilities
- Helps improve compatibility with other UNIX-like systems

Mac OS x

- Mac OS X is also UNIX-based
- Like OpenIndiana, it follows UNIX standards
- Shows how different systems can follow the same standard but have different designs

Solaris

- Solaris is the main base of OpenIndiana
- OpenIndiana is derived from OpenSolaris
- Many features like ZFS and system structure come from Solaris

Limits

- OpenIndiana follows system limits defined by UNIX standards
- These include limits on files, processes, and memory usage
- Helps maintain system stability

Options

- OpenIndiana supports optional UNIX features depending on implementation
- Some advanced features like ZFS and virtualization support are included
- Provides flexibility to users

Primitive system data types

- OpenIndiana uses standard data types defined by C and POSIX
- Ensures compatibility with UNIX programs
- Important for system-level programming

Conflict between standards

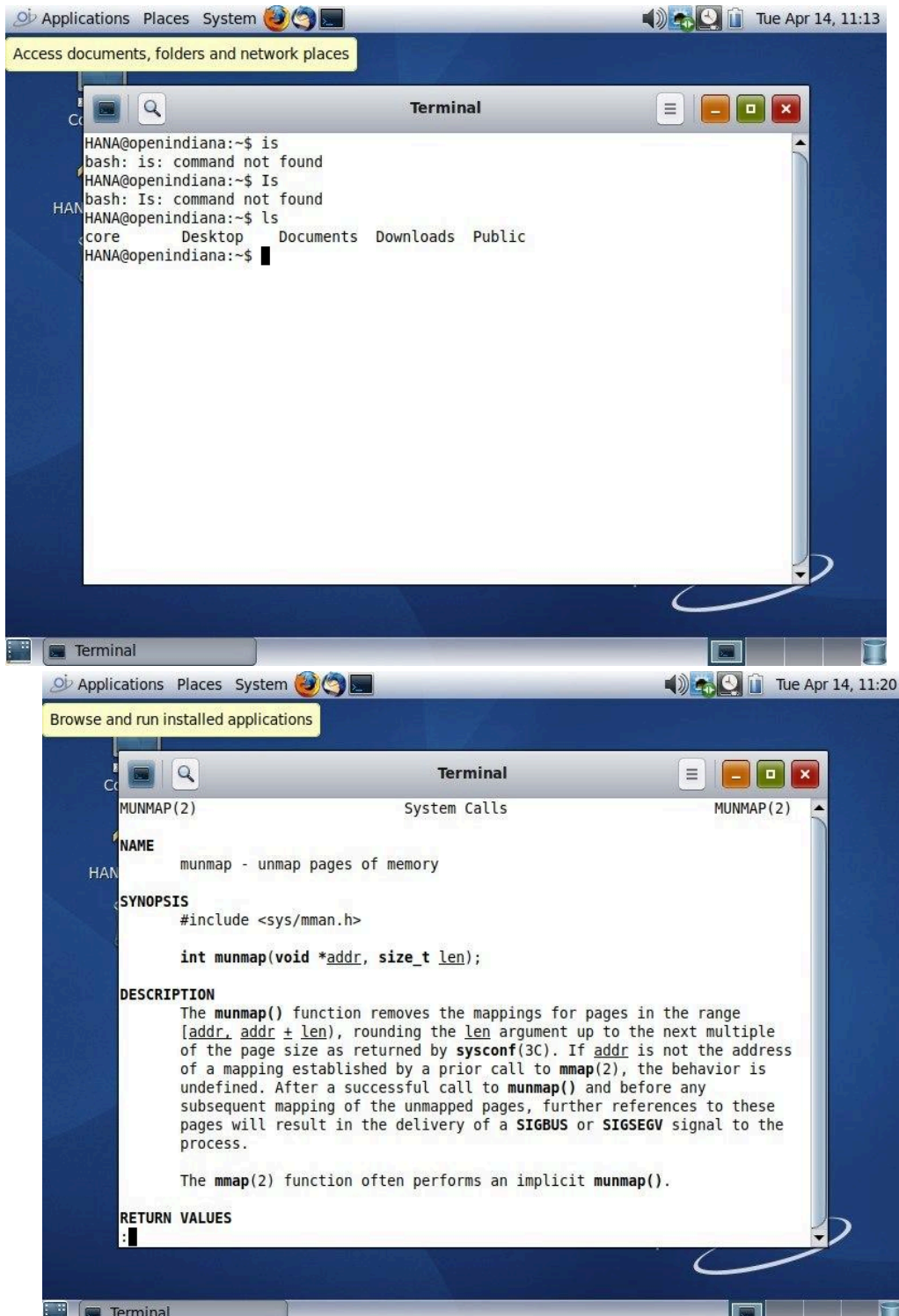
- Sometimes ISO, POSIX, and other standards may have differences

- OpenIndiana tries to follow POSIX while maintaining Solaris features
- Developers must handle these differences carefully

11. System call implementation

System calls are the interface between user programs and the operating system. In OpenIndiana, system calls allow applications to request services from the kernel, such as file handling, process control, and memory management.

In OpenIndiana, system calls are implemented in a layered way. When a user program wants to perform an operation (for example, reading a file), it does not directly access the hardware. Instead, it sends a request to the operating system through a system call.



```
mouse, instance #0
pit_beep, instance #0
display, instance #0
pci8086,1e, instance #0
pci80ee,cafe (driver not attached)
pci8086,0, instance #0
pci106b,3f, instance #0
input, instance #0
pci8086,7113 (driver not attached)
pci8086,265c, instance #0
pci8086,2829, instance #0
disk, instance #0
cdrom, instance #1
fw, instance #0
cpu (driver not attached)
cpu (driver not attached)
sb, instance #1
used-resources (driver not attached)
iscsi, instance #0
agpgart, instance #0 (driver not attached)
options, instance #0
pseudo, instance #0
xsvc, instance #0
HANA@openindiana:~$
```

```
mouse, instance #0
pit_beep, instance #0
display, instance #0
pci8086,1e, instance #0
pci80ee,cafe (driver not attached)
pci8086,0, instance #0
pci106b,3f, instance #0
input, instance #0
pci8086,7113 (driver not attached)
pci8086,265c, instance #0
pci8086,2829, instance #0
disk, instance #0
cdrom, instance #1
fw, instance #0
cpu (driver not attached)
cpu (driver not attached)
sb, instance #1
used-resources (driver not attached)
iscsi, instance #0
agpgart, instance #0 (driver not attached)
options, instance #0
pseudo, instance #0
xsvc, instance #0
HANA@openindiana:~$ vmstat
```



```

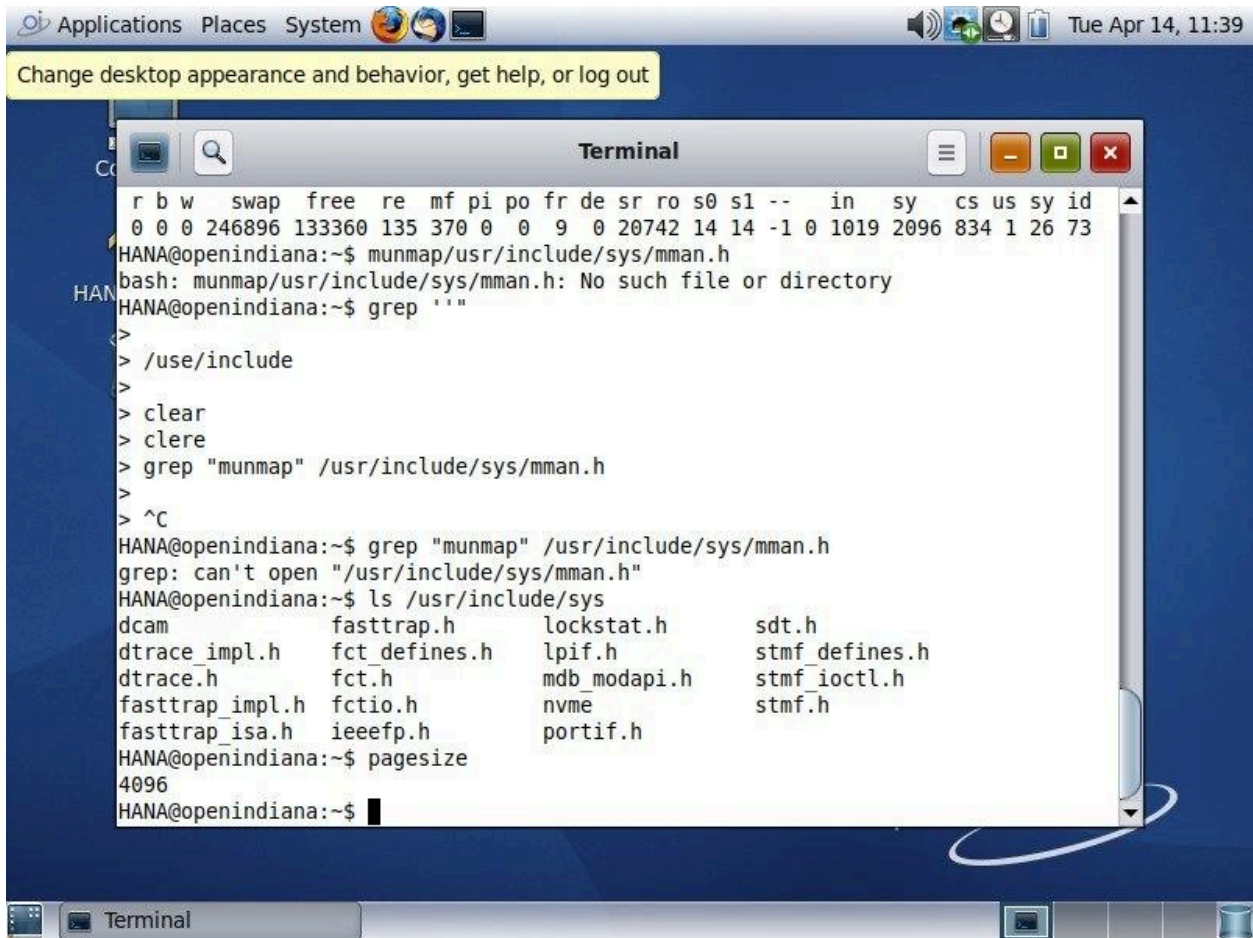
pci80ee,cafe (driver not attached)
pci8086,0, instance #0
pci106b,3f, instance #0
    input, instance #0
pci8086,7113 (driver not attached)
pci8086,265c, instance #0
pci8086,2829, instance #0
    disk, instance #0
    cdrom, instance #1
fw, instance #0
    cpu (driver not attached)
    cpu (driver not attached)
    sb, instance #1
used-resources (driver not attached)
iscsi, instance #0
agpgart, instance #0 (driver not attached)
options, instance #0
pseudo, instance #0
xsvc, instance #0
HANA@openindiana:~$ vmstat
kthr      memory          page            disk           faults        cpu
 r  b w   swap free  re  mf pi po fr de sr ro s0 s1 --  in  sy  cs us sy id
 0  0  0 246896 133360 135 370 0  0  9  0 20742 14 14 -1 0 1019 2096 834 1 26 73
HANA@openindiana:~$

```

```

HANA@openindiana:~$ vmstat
kthr      memory          page            disk           faults        cpu
 r  b w   swap free  re  mf pi po fr de sr ro s0 s1 --  in  sy  cs us sy id
 0  0  0 246896 133360 135 370 0  0  9  0 20742 14 14 -1 0 1019 2096 834 1 26 73
HANA@openindiana:~$ munmap/usr/include/sys/mman.h
bash: munmap/usr/include/sys/mman.h: No such file or directory
HANA@openindiana:~$ grep ''
>
> /usr/include
>
> clear
> clere
> grep "munmap" /usr/include/sys/mman.h
>
> ^C
HANA@openindiana:~$ grep "munmap" /usr/include/sys/mman.h
grep: can't open "/usr/include/sys/mman.h"
HANA@openindiana:~$ ls /usr/include/sys
dcam          fasttrap.h    lockstat.h    sdt.h
dtrace_impl.h fct_defines.h lpif.h         stmf_defines.h
dtrace.h      fct.h         mdb_modapi.h  stmf_ioctl.h
fasttrap_impl.h fctio.h       nvme           stmf.h
fasttrap_isa.h ieeeep.h      portif.h
HANA@openindiana:~$

```




```

00007FFFAF2E0000    1524K r-x-- /lib/amd64/libc.so.1
00007FFFAF46D000     48K rw--- /lib/amd64/libc.so.1
00007FFFAF479000     16K rw--- /lib/amd64/libc.so.1
00007FFFAF48F000      4K rwxS- [ anon ]
00007FFFAF4A0000     64K rwx-- [ anon ]
00007FFFAF4C0000    128K rwx-- [ anon ]
00007FFFAF4EB000     80K r---- /usr/lib/locale/en_US.UTF-8/LC_COLLATE/LCL_D
ATA
00007FFFAF500000     24K rwx-- [ anon ]
00007FFFAF510000      4K rwx-- [ anon ]
00007FFFAF520000      4K rwx-- [ anon ]
00007FFFAF530000      4K rwx-- [ anon ]
00007FFFAF540000      4K rwx-- [ anon ]
00007FFFAF550000      4K rw--- [ anon ]
00007FFFAF560000      4K rw--- [ anon ]
00007FFFAF570000      4K rwx-- [ anon ]
00007FFFAF580000      4K rwx-- [ anon ]
00007FFFAF590000      8K r--S- [ comm ]
00007FFFAF599000    324K r-x-- /lib/amd64/ld.so.1
00007FFFAF5FA000     12K rwx-- /lib/amd64/ld.so.1
00007FFFAF5FD000      8K rwx-- /lib/amd64/ld.so.1
00007FFBFFFA000     24K rw--- [ stack ]
total    4588K
HANA@openindiana:~$

```

12. Over all personal experience

This project was about installing and using OpenIndiana in a virtual machine using Oracle VM VirtualBox. It helped to understand how operating systems are installed, configured, and managed in a safe virtual environment.

During the work, I learned basic system setup, virtualization concepts, and how UNIX-based systems operate. I also faced and solved common installation issues, which improved my practical skills.

OpenIndiana uses the ZFS filesystem and follows UNIX standards, making it stable and reliable for learning and system practice.

13. References

- OpenIndiana Official Documentation – <https://www.openindiana.org>
- Oracle VM VirtualBox User Guide – <https://www.virtualbox.org/manual>

- [UNIX System and POSIX Standards Overview – IEEE Standards Association](#)